

Tobby Meng

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EDUCATION

University of Waterloo

Waterloo, ON

BASc in Electrical Engineering

EXPERIENCE

Software Engineering Intern

September 2026 – December 2026

Ontario Ministry of Public and Business Service Delivery and Procurement

North York, Ontario

- Incoming placement in CSIMB branch.

Software Engineering Intern

January 2026 – April 2026

Qwhery Inc.

Burlington, Ontario

- Built full-stack spatial data platform with a FastAPI backend and interactive dashboard frontend; cut API response latency by 15% through async request handling and connection pooling.
- Designed API schemas with Pydantic validation to catch unwanted inputs across a GIS-to-JSON pipeline; showed results in a responsive dashboard UI with real-time data updates.
- Applied AOT compilation and memory profiling to reduce frontend load times on data-heavy dashboard views.
- Shipped production UI components on agile sprint schedules using Azure DevOps.

Mathematics Teacher

February 2024 – June 2025

Crania Schools

Waterloo, Ontario

- Taught competition math to 20+ students using adaptive curricula targeted at individual skill gaps.
- Built custom problem sets based on diagnostic data, accelerating student core subject progress by 25%.

PROJECTS

Autonomous Infrastructure Agent | Python, gRPC, Docker, Redis

2026

- Built an automated debugging agent that listens to server log streams via gRPC, identifies crash errors, and generates bug fixes automatically.
- Ran generated code fixes inside temporary, isolated Docker containers so untrusted code could be compiled and tested safely before hitting production.
- Implemented a safety checker to block dangerous terminal commands, successfully auto-fixing microservice crashes in under 12 seconds.

RAG Knowledge Base & Search Engine | Python, FastAPI, React, PostgreSQL, pgvector

2026

- Built a search tool that allows users to upload PDF/Markdown documents and ask questions about their content in plain English.
- Stored document embeddings in PostgreSQL to search texts by meaning rather than exact keyword matches, yielding results in under 20ms.
- Connected a React UI to FastAPI via WebSockets to display AI responses word-by-word alongside exact source highlights in the document.
- Used Redis to process multiple document uploads in the background so the user interface never freezes during large file transfers.

AI-Assisted Delirium Detection System | Arduino, Python

November 2025

- Designed a \$50 hardware device using an Arduino microphone circuit that records patient speech to screen for clinical delirium.
- Wrote a Python service that cleans the raw analog audio, converts it to standard .wav files, and sends transcripts to GPT for medical evaluation.
- Delivered full diagnostic results to a clinician's screen in under 30 seconds during test trials.

TECHNICAL SKILLS

Languages & Web: Python, C/C++, TypeScript, JavaScript, SQL, React, Next.js, Angular, HTML/CSS

Backend & Systems: FastAPI, REST APIs, gRPC, Pydantic, Node.js, Docker, Linux, Arduino

Databases & Tools: PostgreSQL, Redis, MongoDB, Git, Azure DevOps